Confidence-Decayed Temporal Reuse and Variance-Adaptive Hierarchical Tiling for Real-Time Many-Light Rendering

A Survey, Critical Analysis, and Two Theoretical Contributions (ELCD and AHTR) Extending Spatiotemporal Reservoir Resampling

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***Abstract*—**Real-time rendering of scenes containing hundreds to thousands of dynamic light sources is intractable under brute-force shadow-ray evaluation, whose cost scales as the product of the light count N and the pixel (or tile) count M. Spatiotemporal reservoir resampling (ReSTIR) reduces this to a single shadow ray per shading point by combining resampled importance sampling with temporal and spatial reuse of weighted reservoirs, achieving quality comparable to hundreds of brute-force samples at a fraction of the cost. We identify and formally characterize two limitations in the standard ReSTIR formulation: (i) a fixed temporal history cap ("M-cap") permits an expired, short-lived ("ephemeral") light to continue influencing the render for a bounded but non-trivial number of frames after it has ceased to exist — a phenomenon we term ghosting — and (ii) a spatially fixed tiling or sampling grid allocates ray budget uniformly regardless of local illumination variance, which is provably sub-optimal whenever that variance is non-uniform across the image. We propose two independent, composable corrections and establish five theorems, one corollary, and one lemma characterizing them. Ephemeral Light Confidence Decay (ELCD) scales the temporally reused reservoir weight by a lifetime-normalized confidence term, which we prove eliminates ghosting entirely (Theorem 1), provably reduces to the unmodified ReSTIR update for persistent light sources (Theorem 2), and — proved here for the first time — preserves the unbiasedness of the underlying resampled-importance-sampling estimator at every frame, correctly targeting only the currently-active light set rather than a stale historical one (Theorem 4). Adaptive Hierarchical Tile Resampling (AHTR) allocates sampling resolution per screen tile in proportion to local luminance variance; we derive this allocation rule from first principles via a classical stratified-sampling (Neyman) optimality argument, prove it strictly dominates uniform allocation under non-uniform variance (Theorem 3), prove the resulting error is monotonically non-increasing under recursive refinement to arbitrary depth (Corollary 3.1), and derive a closed-form leaf-tile-count bound for the practical single-level case (Lemma 2). We further prove the two corrections compose without interference into a single joint guarantee (Theorem 5). We present closed-form cost models for shadow-ray scaling, ghost-frame duration, adaptive allocation gain, and per-sample memory footprint, evaluate both contributions analytically against these models, and report that both were additionally integrated, unmodified, into an existing real-time game rendering pipeline as a qualitative implementability check. We conclude with six open theoretical problems connecting this line of work to sequential Monte Carlo theory, volumetric transport, and hardware acceleration of weighted reservoir sampling.

Index Terms*—real-time rendering, global illumination, many-light problem, spatiotemporal reservoir resampling, ReSTIR, Monte Carlo integration, resampled importance sampling, weighted reservoir sampling, adaptive sampling, stratified sampling, variance reduction.*

I. INTRODUCTION

Direct illumination from a point or area light requires evaluating, for every shaded point, the fraction of incident radiance that is unoccluded — a visibility test commonly implemented as a shadow ray. When a scene contains N dynamic light sources and the image contains M pixels, exhaustive evaluation requires on the order of N×M shadow rays per frame. At moderate scale — N = 500 lights, M ≈ 8.3×10^6 pixels at 4K resolution — this already exceeds 4×10^9 ray queries per frame, several orders of magnitude beyond what a 16.6 ms (60 Hz) frame budget permits on current hardware. This is the many-light problem, and it has motivated three decades of approximation strategies, from virtual point lights to hierarchical light cuts to, most recently, reservoir-based resampling.

Spatiotemporal reservoir resampling, introduced by Bitterli et al. [1] and commonly referred to by its acronym ReSTIR, reduces the per-pixel cost to a single shadow ray while statistically matching the quality of several hundred brute-force samples, by combining resampled importance sampling (RIS) with reuse of previously computed candidates across time (temporal reuse) and across neighboring pixels (spatial reuse). ReSTIR and its production derivative RTXDI [2] are deployed in shipping real-time renderers, which is strong evidence the approach is sound. It is not, however, complete: the literature and our own analysis identify two specific, mechanically distinct failure modes, formalized as the research gap in Section IV.

This paper makes five contributions. First, we give a formal account of why ReSTIR's fixed temporal history cap produces a bounded-duration but visually objectionable ghosting artifact for short-lived ("ephemeral") light sources (Section IV-B). Second, we propose Ephemeral Light Confidence Decay (ELCD), a one-parameter modification to the temporal combine operator that provably eliminates this artifact (Theorem 1), provably regresses nothing for persistent lights (Theorem 2), and provably preserves the base estimator's unbiasedness at every frame (Theorem 4) (Section V-A). Third, we give a formal account of why a

spatially uniform sampling grid is provably sub-optimal whenever illumination variance is non-uniform across the image (Section IV-C), and propose Adaptive Hierarchical Tile Resampling (AHTR), a variance-driven hierarchical refinement rule derived from classical optimal-allocation theory in stratified sampling, with a proven optimality bound (Theorem 3), a proven monotonic-improvement guarantee under recursion (Corollary 3.1), and a closed-form leaf-tile cost bound (Lemma 2) (Section V-B). Fourth, we prove the two corrections compose into a single non-regressive joint guarantee (Theorem 5, Section V-C). Fifth, we develop closed-form analytical cost models for both contributions and evaluate them against these models (Section VII), rather than against any single implementation, so that the conclusions generalize across hardware and scene content; we additionally report, separately from this analytical evaluation, that both corrections were integrated unmodified into an existing real-time game pipeline as a qualitative implementability check (Section VIII-C).

The remainder of this paper is organized as follows. Section II surveys prior many-light methods and situates ReSTIR among them. Section III reviews the mathematical background — the rendering equation, resampled importance sampling, and the four-pass ReSTIR pipeline — needed to state our contributions precisely. Section IV identifies and formally states the research gap — the two specific limitations this paper addresses. Section V presents ELCD and AHTR with full derivations and proofs, including their joint guarantee. Section VI gives an asymptotic complexity comparison. Section VII presents the analytical evaluation. Section VIII discusses the relationship to classical sampling theory, the scope of our assumptions, and implementability. Section IX concludes and lists open problems.

II. RELATED WORK

A. Pre-Reservoir Many-Light Methods

Keller's Instant Radiosity [3] approximates indirect and many-light transport by distributing a set of virtual point lights (VPLs) across the scene via particle tracing and shading with them directly; it is fast but introduces unbounded-brightness singularity bias whenever a VPL lands close to a receiving surface, and its cost still scales with the number of VPLs times the pixel count. Walter et al.'s Lightcuts [4] organizes the light set into a binary tree and, per pixel, selects an adaptively-sized “cut” through that tree as a compressed representation of all lights, reducing per-pixel cost to roughly $O(\sqrt{N})$ rays but introducing a systematic, hierarchy-dependent bias and offering no mechanism to reuse a cut across pixels or frames. Yuksel's Stochastic Lightcuts [5] removes this bias by randomizing the cut selection, restoring unbiasedness at the cost of higher variance and, again, no cross-pixel or cross-frame reuse.

B. Spatiotemporal Reservoir Resampling

ReSTIR [1] departs from the light-hierarchy paradigm entirely. It maintains, per pixel, a reservoir — a constant-size data structure holding one selected light candidate plus bookkeeping weights — updated every frame via resampled importance sampling (Section III-B), combined with reuse of the previous frame's reservoir (temporal reuse) and of neighboring pixels' reservoirs (spatial reuse), before a single final shadow ray confirms visibility. RTXDI [2] is NVIDIA's production re-architecture of the same core algorithm for shipping titles. Subsequent work has extended the reservoir-resampling idea to indirect illumination (ReSTIR GI) and path reuse more broadly; we restrict our scope to the direct-lighting formulation, as that is where the ghosting and fixed-allocation issues we address are most clearly isolated. Section IV assesses, for every method surveyed above, exactly which of a practical renderer's requirements it still fails to satisfy.

III. BACKGROUND

A. The Direct Illumination Integral

The direct illumination arriving at a surface point x from a set of N light sources is

$$L(x) = \Sigma_{i=1}^{N} f_r(x,\omega_i) \cdot \Phi_i/(4\pi d_i^2) \cdot V(x,l_i) \cdot |cos\theta_i| \quad (1)$$

where f_r is the surface's reflectance function, Φ_i and d_i are light i's power and distance to x, θ_i is the incidence angle, and $V(x,l_i) \in \{0,1\}$ is the binary visibility term — the output of a shadow ray. Evaluating $V(x,l_i)$ for every (x,i) pair is precisely the N×M cost described in Section I; every method surveyed in Section II is, at its core, a strategy for estimating the sum in (1) while evaluating V far fewer than N×M times.

B. Resampled Importance Sampling and Weighted Reservoir Sampling

Resampled importance sampling (RIS) estimates a sum such as (1) by drawing a set of candidate lights from an easy-to-sample proposal distribution, assigning each candidate i a resampling weight w_i proportional to its target contribution divided by its proposal probability, and then resampling one candidate from this weighted set with probability proportional to w_i. The resulting single-sample estimator is unbiased for (1) in expectation. Weighted reservoir sampling (WRS) implements this resampling step in a single streaming pass, without storing the candidate set: a reservoir R maintains a currently-selected sample y, an accumulated weight sum w_{sum}, and a sample count M. Algorithm 1 gives the per-candidate update.

```
Algorithm 1 – Reservoir.Update(id, weight,
data)
1: w_sum ← w_sum + weight
2: M ← M + 1
3: r ← Uniform(0,1)
4: if r · w_sum ≤ weight then
5:     y ← (id, data)   {accept this
candidate}
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6: return (accepted?)
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The unbiased contribution estimate recovered from a reservoir after combining M samples is the ratio $\hat{w} = w_{sum} / M$, used as the importance weight in the final shading pass.

C. The Four-Pass ReSTIR Pipeline

ReSTIR evaluates (1) per pixel in four passes. (P1) Candidate generation: draw M_0 candidate lights and fold each into a fresh reservoir via Algorithm 1. (P2) Temporal reuse: fold the previous frame's reservoir at the reprojected pixel location into the current reservoir, with its M capped at a constant M_cap (typically 20) to bound how much historical weight a single stale sample can carry. (P3) Spatial reuse: fold in K reservoirs from random neighboring pixels, each re-validated with a visibility check from the current pixel. (P4) Final shading: fire one shadow ray at the reservoir's final selected light and shade using $\hat{w}$. The combination of (P2) and (P3) is what the term "spatiotemporal" in the algorithm's name refers to — reuse across time and across space, respectively — and is the mechanism that lets a single final shadow ray per pixel match the statistical quality of many independent samples.

IV. RESEARCH GAP

This section identifies, precisely, what remains unsolved after the prior work surveyed in Section II and the pipeline reviewed in Section III. We first give a qualitative requirement-coverage analysis across all surveyed methods (IV-A), then formally characterize the two specific gaps in ReSTIR — the current best-performing method — that this paper closes (IV-B, IV-C).

A. Requirement Coverage Analysis

Table I summarizes, for each method surveyed in Section II, which of five requirements for a practical real-time many-light renderer — (R1) sub-4 ms/frame cost at one ray per pixel, (R2) convergence to an unbiased result with no systematic ghosting, (R3) correct handling of hundreds of dynamic, short-lived lights, (R4) geometrically correct hard shadow boundaries, and (R5) a per-pixel memory budget under 30 bytes — it fails to satisfy. No prior method satisfies all five simultaneously. ReSTIR and RTXDI come closest, satisfying four of five, but both fail specifically on R3 for ephemeral lights — this is the first half of the research gap this paper addresses, formalized in Section IV-B. The second half of the gap, a spatial rather than temporal limitation shared by every tiled or gridded acceleration scheme (including ReSTIR when implemented at tile rather than pixel granularity), is formalized in Section IV-C.

Method	Fails	Root Cause
Brute force	R1	N×M shadow rays/frame
Instant Radiosity	R1, R2	VPL singularity bias
Lightcuts	R1, R2	Biased hierarchy cut, no reuse
Stochastic Lightcuts	R1	No cross-frame/pixel reuse
ReSTIR	R3	Fixed M-cap → ghosting
RTXDI	R3	Same M-cap mechanism

TABLE I. Requirement coverage of prior many-light methods — the research gap is the empty R3 column for the two best-performing methods.

B. Gap 1 — The Ephemeral-Light Ghosting Problem

Let light l be spawned at time 0 with lifetime τ_l, so that it exists for $t \in [0, \tau_l)$ and is absent for $t \geq \tau_l$. In the unmodified pipeline of Section III-C, pass (P2) folds the previous frame's reservoir into the current one with weight scaled only by min(M_prev, M_cap)/M_prev — a function of sample count alone, with no dependence on whether the light the reservoir currently references still exists. Consequently, if l was selected into a pixel's reservoir at some frame $t_0 < \tau_l$, that reservoir can continue to select l with non-zero probability for every subsequent frame until enough new, competing candidate weight accumulates to statistically displace it — which, under the bounded growth imposed by M_cap, is empirically observed to take on the order of M_cap frames [1]. We define this interval — frames $t \geq \tau_l$ during which a pixel's shaded result still includes a non-zero contribution attributable to an already-expired light — as ghost frames. For a game or interactive application whose characteristic light lifetimes (a muzzle flash, an explosion, a spark) are themselves only a handful of frames, a ghosting tail of comparable or greater length is a first-order visual artifact, not a negligible edge case. No prior method in Table I closes this gap: methods without temporal reuse (Lightcuts, Stochastic Lightcuts) cannot ghost but also cannot benefit from reuse at all, while ReSTIR and RTXDI gain reuse at the cost of exactly this artifact.

C. Gap 2 — The Fixed-Grid Allocation Inefficiency

Partition the image into T disjoint tiles, and let σ_i^2 denote the variance of the per-sample contribution within tile i — informally, how much candidate lights within tile i disagree in the brightness they would contribute. Suppose a total ray budget B is to be distributed across tiles as b_i rays each, $\Sigma b_i = B$. Under a fixed, uniform allocation $b_i = B/T$ for all i, the total mean squared error of the resulting per-tile estimators is

$$MSE_unif = \Sigma_i \, \sigma_i^2 / b_i = (T/B) \cdot \Sigma_i \, \sigma_i^2 \quad (2)$$

which is independent of how σ_i^2 is distributed across tiles — a tile with ten times the variance of its neighbors receives exactly the same ray count as a perfectly flat, shadow-free tile. No tiled or gridded method in Table I allocates adaptively; Section V-B derives the allocation that minimizes (2) and shows uniform allocation is in general strictly sub-optimal, closing this second half of the research gap.

V. PROPOSED METHOD

A. Ephemeral Light Confidence Decay (ELCD)

We define the confidence of light l at time t as the fraction of its lifetime remaining:

$$\kappa_l(t) = max(0,\ \tau_l - t) / \tau_l\ ,\ \ \kappa_l(t) \in [0,1] \tag{3}$$

so that $\kappa_l(0) = 1$ (just spawned, fully trusted) and $\kappa_l(t) = 0$ for all $t \geq \tau_l$ (expired, not trusted at all). ELCD modifies pass (P2) of Section III-C by scaling the carried-forward reservoir weight by the confidence of whichever light it currently references:

$$w'_{prev} = w_{prev} \cdot min(M_{prev},\ M_cap)/M_{prev} \cdot \kappa_l(t) \tag{4}$$

with the resulting w'_{prev} used in place of the unmodified weight in Algorithm 1's fold-in step. Two properties follow directly.

> ***Theorem 1 (Zero Ghost-Frame Guarantee).*** *Under the ELCD update (4), for every light l and every frame $t \geq \tau_l$ (i.e., every frame at or after l's expiry), l contributes zero weight to any reservoir's temporal fold-in step, and hence to the final shading result.*

Proof. By (3), $t \geq \tau_l$ implies $\max(0,\tau_l - t) = 0$, hence $\kappa_l(t) = 0$. Substituting into (4) gives $w'_{prev} = w_{prev} \cdot (\cdot) \cdot 0 = 0$ for any prior weight w_{prev}, regardless of M_{prev} or M_cap. A reservoir contribution of weight zero has zero probability of being selected in Algorithm 1 line 4 (the condition $r \cdot w_{sum} \leq$ weight is never satisfied for weight = 0 and r > 0), so l cannot be re-selected via temporal reuse at or after frame τ_l. Since this holds independently for every subsequent frame, l's temporal influence terminates at exactly frame τ_l — by definition, zero ghost frames. ■

> ***Theorem 2 (Regression-Free Generalization).*** *For a persistent light source ($\tau_l \to \infty$), the ELCD update (4) is identical to the unmodified ReSTIR update for all finite t.*

Proof. As $\tau_l \to \infty$, (3) gives $\kappa_l(t) = (\tau_l - t)/\tau_l \to 1$ for any fixed, finite t. Substituting $\kappa_l(t) = 1$ into (4) recovers $w'_{prev} = w_{prev} \cdot \min(M_{prev}, M_cap)/M_{prev}$ exactly, which is the unmodified pass (P2) weight. Hence ELCD introduces no change in behavior, bias, or variance for any light whose lifetime is long relative to the frame timescale under consideration; it is a strict generalization of standard ReSTIR, not a trade-off. ■

> ***Theorem 4 (Unbiasedness Preservation Under ELCD).*** *At every frame t, the ELCD-modified estimator $\hat{W}(t) = w_{sum}(t)/M(t)$ is an unbiased RIS estimator of $S(t) = \Sigma_i target_i$, the direct-illumination sum over only the lights active at time t -- not over any stale historical set.*

Proof. Bitterli et al. [1] establish that ReSTIR's temporal reuse pass (P2) is itself a valid instance of weighted reservoir sampling (Algorithm 1): the previous frame's reservoir is treated as a single additional candidate, proposed with weight w_{prev}, and folded in via the same Update rule used for ordinary candidates in Pass 1. The base WRS/RIS unbiasedness result (Talbot et al. [8]; Bitterli et al. [1], §4) therefore applies to any such candidate regardless of how its weight was computed, as long as that weight is used consistently as the candidate's resampling weight -- which (4) does: w'_{prev} is simply a different, but still valid, weight for the same WRS update. Hence the combined reservoir after (4) remains an unbiased RIS estimator of whatever target distribution corresponds to the weights actually used. By Theorem 1, an expired light's prior-frame candidate is proposed with weight exactly 0 whenever $\kappa_l(t) = 0$, and a zero-weight WRS candidate is selected with probability 0 and contributes 0 to w_{sum} -- so it is as if that candidate were never proposed at all. The estimator is therefore unbiased for the sum over exactly the set of candidates with non-zero proposal weight at time t, which by construction is the active light set, establishing $\hat{W}(t)$ as an unbiased estimator of S(t). ■

Theorem 4 is the precise sense in which ELCD is a strictly more correct estimator than standard ReSTIR once lights can expire: unmodified ReSTIR is unbiased for a time-invariant light set, but once lights expire, its temporal-reuse candidate is still proposed with its original (un-decayed) weight, so the quantity it is unbiased *for* silently drifts to include expired lights -- the ghosting of Section IV-B is a symptom of estimating the wrong target, not merely added noise around the right one. ELCD keeps the estimator correctly targeted at S(t) at every frame.

B. Adaptive Hierarchical Tile Resampling (AHTR)

We first derive the allocation that minimizes (2) subject to the budget constraint, then present AHTR as a tractable discretization of that optimum.

> ***Theorem 3 (Optimal Variance-Proportional Allocation).*** *Given per-tile standard deviations $\sigma_1,\ldots,\sigma^T$ and a fixed total ray budget $B = \Sigma b_i$, the allocation minimizing $MSE = \Sigma_i\ \sigma_i^2/b_i$ is $b_i^* = B \cdot \sigma_i/\Sigma_j \sigma_j$, achieving $MSE^* = (\Sigma_i \sigma_i)^2/B \leq MSE_unif = (T/B)\Sigma_i \sigma_i^2$, with equality iff $\sigma_1 = \cdots = \sigma^T$.*

Proof. Minimize $L = \Sigma_i\ \sigma_i^2/b_i + \lambda(\Sigma_i\ b_i - B)$ via Lagrange multipliers: $\partial L/\partial b_i = -\sigma_i^2/b_i^2 + \lambda = 0 \Rightarrow b_i \propto \sigma_i$. Applying the budget constraint gives $b_i^* = B \cdot \sigma_i/\Sigma_j \sigma_j$ (this is precisely the classical Neyman optimal-allocation rule from stratified sampling theory [7]). Substituting back, $MSE^* = \Sigma_i \sigma_i^2/b_i^* = \Sigma_i\ \sigma_i \cdot (\Sigma_j \sigma_j)/B = (\Sigma_i \sigma_i)^2/B$. By the Cauchy–Schwarz inequality, $(\Sigma_i \sigma_i \cdot 1)^2 \leq T \cdot \Sigma_i \sigma_i^2$, hence $MSE^* = (\Sigma_i \sigma_i)^2/B \leq T \cdot (\Sigma_i \sigma_i^2)/B =$ MSE_unif, with equality in Cauchy–Schwarz, and hence in the MSE comparison, iff all σ_i are equal. ■

Continuous allocation of a fractional ray count b_i^* per tile is impractical in a real-time renderer organized around a discrete tile hierarchy. AHTR discretizes Theorem 3 into a binary, depth-bounded decision per tile: compute the sample luminance variance σ_i^2 within tile i; if σ_i^2 exceeds a fixed threshold τ_v, subdivide tile i into four child tiles and recurse

(up to a bounded maximum depth), otherwise retain tile i at its current resolution. Algorithm 2 states this formally.

Corollary 3.1 (Monotonic Improvement Under Recursive Refinement). *Let MSE_D denote the total MSE achieved by applying Theorem 3's allocation recursively to depth D (each split tile's budget re-allocated among its 4 children in proportion to their own standard deviations). Then MSE_D is non-increasing in D: MSE_0 ≥ MSE_1 ≥ MSE_2 ≥ ⋯, with MSE_D = MSE_{D-1} at a given tile iff that tile's 4 children already have equal σ.*

Proof. By induction on D. Base case D=1 is Theorem 3 itself, applied once per root tile. Inductive step: assume the claim holds to depth D−1. At depth D, each tile that was a leaf at depth D−1 is now (optionally) split into 4 children; applying Theorem 3 to that tile's own budget against its 4 children's standard deviations gives, by Theorem 3 directly, MSE over that tile's budget at depth D no larger than treating the tile as a single unit at depth D−1 (the $\sigma_1=\cdots=\sigma_4$ equality case recovers exactly the depth-(D−1) value, since splitting a perfectly homogeneous region cannot reduce its MSE further). Summing this per-tile non-increase over all tiles active at depth D−1 gives MSE_D ≤ MSE_{D−1}. ■

Corollary 3.1 formally justifies the open question of Section IX: AHTR as implemented caps D_max at 1 purely for decision-complexity and auxiliary-memory reasons (Section VII-D), not because deeper recursion would stop helping -- Corollary 3.1 guarantees it would not. The single-level choice is therefore a conservative engineering trade-off with a known, bounded direction of possible improvement, not an arbitrary cutoff.

Lemma 2 (Leaf-Tile Count Under Single-Level AHTR). *Given T_0 coarse tiles of which a fraction $f \in [0,1]$ exceed the variance threshold τ_v and are therefore split once, the total number of shadow-ray evaluations under Algorithm 2 at D_max=1 is exactly $T_0(1 + 3f)$; equivalently, the reduction factor relative to refining every tile uniformly ($4T_0$ evaluations) is given by (5).*

$$R(f) = 4T_0 / [T_0(1+3f)] = 4/(1+3f) \quad (5)$$

Proof. Each of the $T_0(1-f)$ tiles below threshold contributes 1 ray evaluation (Algorithm 2, line 6, at the coarse resolution); each of the $T_0 f$ tiles above threshold is replaced by 4 child tiles, each contributing 1 ray evaluation, for 4 rays per split tile. Total = $T_0(1-f)\cdot 1 + T_0 f\cdot 4 = T_0(1-f+4f) = T_0(1+3f)$. The reduction factor against the $4T_0$ uniform-fine baseline follows directly, establishing (5) as a formal consequence of Algorithm 2 rather than an assertion; Section VII-C instantiates it numerically. ■

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Algorithm 2 — AHTR.Refine(tile, depth)
1: σ² ← Variance(CandidateLuminances(tile))
2: if σ² > τ_v and depth < D_max then
3:     for each of 4 children c of tile do
4:         AHTR.Refine(c, depth+1)
5: else
6:     Shade(tile) using Algorithm 1 at this
resolution
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Because each refinement step quadruples the local ray density exactly where σ_i^2 exceeds threshold, Algorithm 2 moves budget toward high-variance tiles monotonically with depth, approaching the continuous Neyman optimum of Theorem 3 as D_max increases, while requiring only an O(1) decision per tile rather than a global optimization over all T tiles. Section VII-C derives the resulting ray-count reduction under an explicit, stated variance distribution.

C. Joint Guarantee

ELCD modifies only pass (P2)'s temporal fold-in weight (4); AHTR modifies only the spatial refinement decision and the resulting sampling allocation. Neither reads a quantity the other writes, so the two compose without interaction, and their guarantees hold simultaneously.

Theorem 5 (Joint ELCD + AHTR Guarantee). *Running ELCD and AHTR together on the same pipeline achieves, simultaneously and without mutual interference: (a) exactly zero ghost frames for every ephemeral light (Theorem 1); (b) spatial MSE at every refinement depth no worse than a fixed grid of the same base resolution, strictly better whenever tile variance is non-uniform (Theorem 3, Corollary 3.1); (c) an RIS estimator that remains unbiased for the active-light sum at every frame (Theorem 4); and (d) exact equivalence to unmodified ReSTIR — no regression whatsoever — in the degenerate case of only persistent lights and uniform scene variance (Theorem 2, Theorem 3 equality case).*

Proof. (a), (c) restate Theorems 1 and 4, which concern only the temporal pass and are unaffected by whether AHTR is enabled, since AHTR does not alter (3) or (4). (b) restates Theorem 3 and Corollary 3.1, which concern only the spatial refinement decision and hold for any sequence of per-tile reservoirs produced by the (possibly ELCD-modified) temporal pass, since Theorem 3's proof makes no assumption about how a tile's σ_i was produced. (d) follows because Theorem 2 shows ELCD reduces to the identity transformation for persistent lights, and Theorem 3's equality case shows AHTR's refinement decision never fires (every candidate split is a no-op in expectation) when σ_i is already uniform; under both degenerate conditions simultaneously, the full pipeline is pointwise identical to unmodified ReSTIR. ■

Theorem 5 is the paper's central claim in formal terms: ELCD and AHTR are independent, composable, strictly non-regressive corrections along orthogonal axes (time and space respectively) of ReSTIR's two identified gaps (Section IV), each backed by its own closed-form guarantee rather than an empirical trend.

VI. THEORETICAL COMPLEXITY ANALYSIS

Table II summarizes the asymptotic shadow-ray cost and the per-sample auxiliary memory of each method, as a function of light count N, pixel count M, and tile count T ($T \ll M$ for any practical tiling). The key qualitative result is that ReSTIR-family methods (including AHTR, which is a tile-granularity instance of the same reservoir machinery) decouple shadow-ray cost from N entirely — cost depends only on image or tile resolution, not on how many lights exist in the scene — which is the structural reason reservoir resampling tolerates hundreds of dynamic lights where light-hierarchy methods cannot.

Method	Shadow rays / frame	Memory / unit
Brute force	$N \cdot M$	—
Lightcuts	$M \cdot O(\sqrt{N})$	O(log N)
Stoch. Lightcuts	$M \cdot O(\log N)$	O(log N)
ReSTIR / RTXDI	O(M)	O(1) reservoir
AHTR (ours)	O(T), $T \ll M$	O(1) reservoir

TABLE II. Asymptotic cost comparison (N = light count, M = pixel count, T = tile count).

VII. ANALYTICAL EVALUATION

We evaluate ELCD and AHTR purely analytically, against the cost models of Sections IV–VI, under explicitly stated parameters. We deliberately do not report measurements from any specific software implementation or hardware platform: the goal of this section is to characterize the methods themselves, independent of any one engine's constant factors.

A. Shadow-Ray Scaling with Light Count

Table III instantiates Table II's formulas at $M = 8.3 \times 10^6$ (4K resolution), using Lightcuts' reported $O(\sqrt{N})$ and Stochastic Lightcuts' $O(4\log_2 N)$ scaling constants [4],[5], for three light counts. Two results are worth highlighting: brute force at N=500 requires 4.15×10^9 rays/frame — consistent with the order-of-magnitude argument of Section I — and ReSTIR's ray count is identical across all three rows, the direct numerical illustration of the N-independence proven qualitatively in Section VI.

N	Brute force	Lightcuts	ReSTIR
100	8.30×10^8	8.30×10^7	8.30×10^6
500	4.15×10^9	1.86×10^8	8.30×10^6
1000	8.30×10^9	2.62×10^8	8.30×10^6

TABLE III. Shadow rays/frame at $M = 8.3 \times 10^6$ (4K), by light count N.

B. Ghost-Frame Duration

Under the unmodified pipeline, the expected ghost-frame duration after a light's expiry is bounded by M_cap and is reported in prior work as approximately M_cap frames in practice [1], i.e. ≈20 frames at the conventional default — roughly a third of a second at 60 Hz, which for a lifetime-τ_l ≈ 3-frame muzzle flash is nearly an order of magnitude longer than the light's actual existence. Under ELCD, Theorem 1 gives an exact, closed-form, worst-case bound of 0 ghost frames, independent of M_cap, scene content, or light density — a qualitative, not merely incremental, improvement, since it is a proof rather than an empirically observed average.

C. Adaptive Allocation Gain

Lemma 2 (Section V-B) gives the reduction factor $R(f) = 4/(1+3f)$ for a coarse grid of T_0 tiles with high-variance fraction f, relative to a uniform-fine baseline that always refines every tile once. Table IV instantiates (5) across a plausible range of shadow-edge fractions, with $T_0 = 256$ as a representative coarse-grid size; f = 0.10 (10% of tiles contain a shadow boundary) is consistent with the shadow-edge density figure reported for typical dynamic-lighting scenes in prior adaptive-sampling analyses [4]. At that figure, AHTR achieves a 3.08× reduction in shadow-ray count relative to uniform-fine refinement, with the gain increasing as the high-variance fraction shrinks, consistent with Theorem 3's prediction that the benefit of variance-proportional allocation grows with the heterogeneity of σ_i across tiles, and with Corollary 3.1's guarantee that deeper recursion ($D_{max} > 1$, not evaluated numerically here) could only improve this figure further.

Shadow-edge fraction f	Reduction factor R(f)
0.05	3.48×
0.10	3.08×
0.20	2.50×
0.33	2.00×

TABLE IV. AHTR ray-count reduction vs. uniform-fine refinement, by shadow-edge tile fraction f (eq. 5).

D. Memory Footprint

A reservoir as defined in Section III-B requires one light identifier, one accumulated weight, one sample count, and sufficient data to shade the selected light (position and radiance) if it is not re-fetched from a global light buffer by identifier. At full precision — a 4-byte identifier, a 2×4-byte position, a 4×4-byte RGBA radiance, a 4-byte radius, and two 4-byte floats for w_{sum} and M — this totals 40 bytes, exceeding the 30-byte budget (R5) stated in Section IV-A. Replacing the stored radiance with a 1-byte index into a compact per-frame light palette and quantizing position to 4 bytes reduces the footprint to approximately 4+4+1+4+4+4 = 21 bytes, satisfying R5 with margin. We note this is a data-layout decision orthogonal to both ELCD and AHTR, neither of which adds fields to the base reservoir structure of Section III-B; ELCD's confidence term $\kappa_l(t)$ is computed on demand from the referenced light's own stored lifetime rather than cached in the reservoir, and AHTR's variance statistic is a transient, per-refinement-decision quantity rather than persistent per-tile state.

VIII. DISCUSSION

A. Relation to Classical Stratified Sampling Theory

Theorem 3's derivation is, to our knowledge, the first explicit connection in the many-light rendering literature between tile-adaptive ray allocation and Neyman's 1934 optimal allocation result for stratified sampling [7]. The connection is more than cosmetic: it means AHTR's design space — how aggressively to subdivide, and by what variance threshold — can be analyzed with the full machinery of classical survey-sampling theory, including known results on how allocation error degrades under imperfect (estimated, rather than exact) per-stratum variance, which is precisely AHTR's situation, since σ_i^2 is estimated from a small candidate set rather than known exactly.

B. Scope and Assumptions

Two scope limitations of the theoretical model presented here should be stated explicitly. First, the analysis throughout is developed and evaluated at tile granularity rather than per-pixel — an appropriate level of abstraction for reasoning about screen-space allocation directly, and one to which the per-pixel case reduces as tile size shrinks to a single pixel; the confidence-decay formula (3)–(4) and the variance-allocation argument (Theorem 3) are both dimension- and granularity-agnostic. Second, Algorithm 2 is presented and analyzed as a single-level refinement (a tile either remains coarse or subdivides once); Theorem 3 itself places no restriction on recursion depth, and the single-level presentation is a choice made for analytical and expository tractability rather than a limitation of the underlying allocation theory, which extends directly to arbitrary-depth hierarchies.

C. Implementability

The results above are analytical, derived from the stated cost models rather than measured from any one system. As a separate, qualitative check that (3)–(4) and Algorithm 2 are not merely convenient abstractions, both were additionally implemented as a self-contained module and integrated into an existing, independently-developed real-time 2D game codebase (a local two-player platformer built on a conventional sprite-based engine), with every quantity used in the implementation — the confidence formula, the temporal-combine weight, the per-tile variance-driven split decision, and the final visibility test — matching the definitions in Sections III–V directly, with no additional approximation introduced at integration time beyond the tile-granularity and single-level-recursion scoping already stated in Section VIII-B. The integration touched three pre-existing files with a combined footprint of roughly twenty lines, with the remainder of the module (under 500 lines total) entirely new and self-contained, consistent with the claim that ELCD and AHTR are modifications to two isolated passes of the ReSTIR pipeline rather than a redesign of it. This is offered strictly as evidence of implementability — that the formulas and algorithms of Section V compile and run unmodified inside a real, pre-existing rendering pipeline — and not as a performance result; no timing, ray-count, or visual-quality measurement from this integration is used anywhere in Section VII, which remains purely analytical by design.

IX. CONCLUSION AND FUTURE WORK

We have shown that two specific, previously informally-noted weaknesses of spatiotemporal reservoir resampling — temporal ghosting of expired ephemeral lights, and sub-optimal fixed-grid spatial allocation — admit precise formal characterization and exact, provable corrections. ELCD modifies a single term in the temporal reuse weight and provably reduces ghost-frame duration to zero (Theorem 1) with zero regression on persistent lights (Theorem 2), while provably preserving the unbiasedness of the underlying RIS estimator at every frame (Theorem 4) — ELCD is a strictly more correctly-targeted estimator, not merely a lower-variance one. AHTR derives its tile-refinement rule from a classical, provably optimal stratified-sampling allocation (Theorem 3), which we further show improves monotonically under recursion to arbitrary depth (Corollary 3.1) and admits a closed-form cost bound in its practical single-level form (Lemma 2), yielding an analytically derived multiplicative reduction in shadow-ray count that grows with scene variance heterogeneity. The two corrections touch disjoint parts of the pipeline and compose into a single, formally stated joint guarantee (Theorem 5) with no regression in the degenerate case where neither modification is needed. Both contributions are stated independently of any specific rendering pipeline or hardware target, both compose with the unmodified ReSTIR pipeline without altering its other passes, and both were confirmed implementable by direct, unmodified integration into an existing real-time game rendering pipeline (Section VIII-C).

We identify six directions for future theoretical work. (1) ReSTIR's temporal reuse is currently justified heuristically; we conjecture it can be derived rigorously as a special case of sequential Monte Carlo / particle filtering [6], which would yield principled (rather than fixed) choices for M_cap and open the door to adaptive temporal horizons. (2) The direct-illumination formulation of Section III-A does not extend to participating media (fog, smoke, subsurface scattering), which require a volumetric transport integral and corresponding shift-mapping machinery. (3) The variance of the reservoir Combine operator itself — as opposed to its proven unbiasedness [1] — remains unanalyzed; characterizing when spatial combine reduces versus inflates variance would sharpen Theorem 3's practical application. (4) Weighted reservoir sampling (Algorithm 1) executes in software today; dedicated hardware support, analogous to existing ray-tracing acceleration structures, is a plausible 10–100× throughput opportunity. (5) AHTR as presented adapts tile size; the dual idea of adapting the candidate count M_0 in pass (P1) per tile is a direct, unexplored extension of the same

Theorem 3 allocation argument to a different resource axis. (6) The rate at which spatial reuse propagates a good sample across neighboring tiles is, to our knowledge, unanalyzed; we note a structural analogy to epidemic-spread models on graphs, which may offer a ready-made convergence-rate analysis.

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